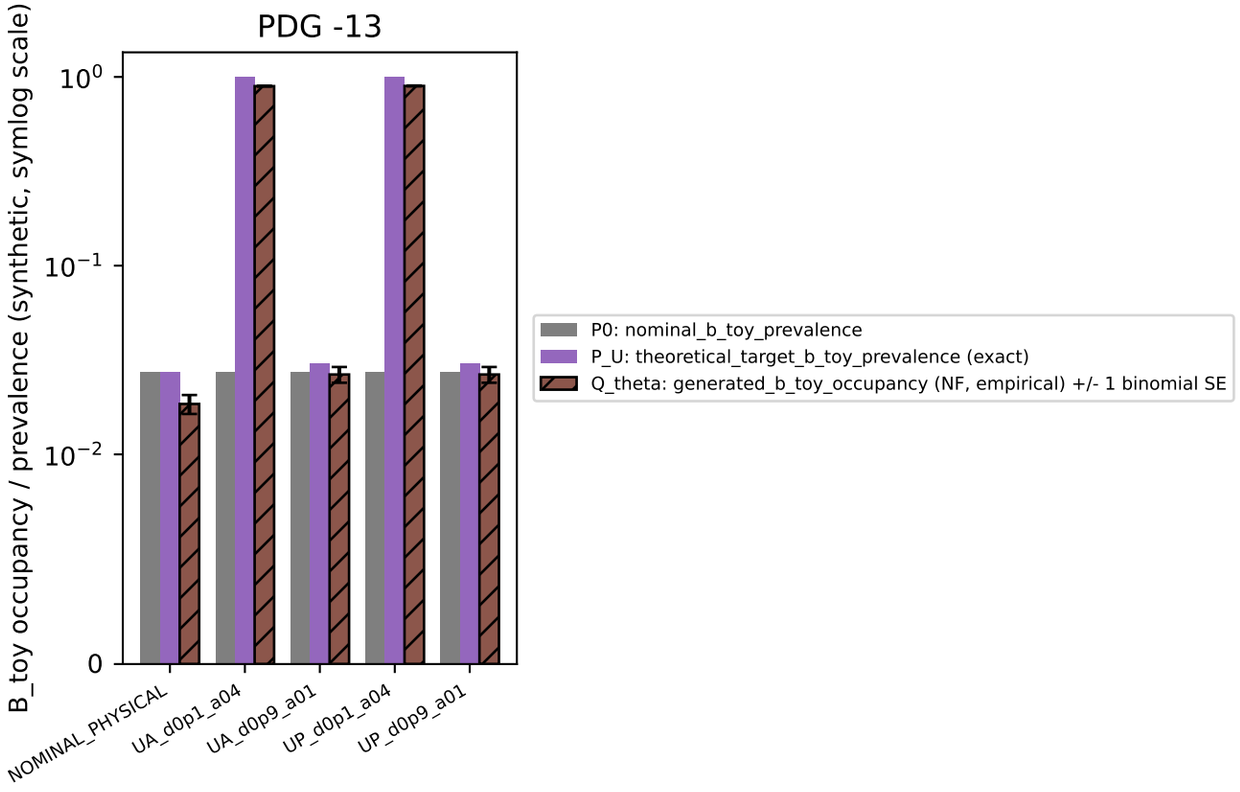
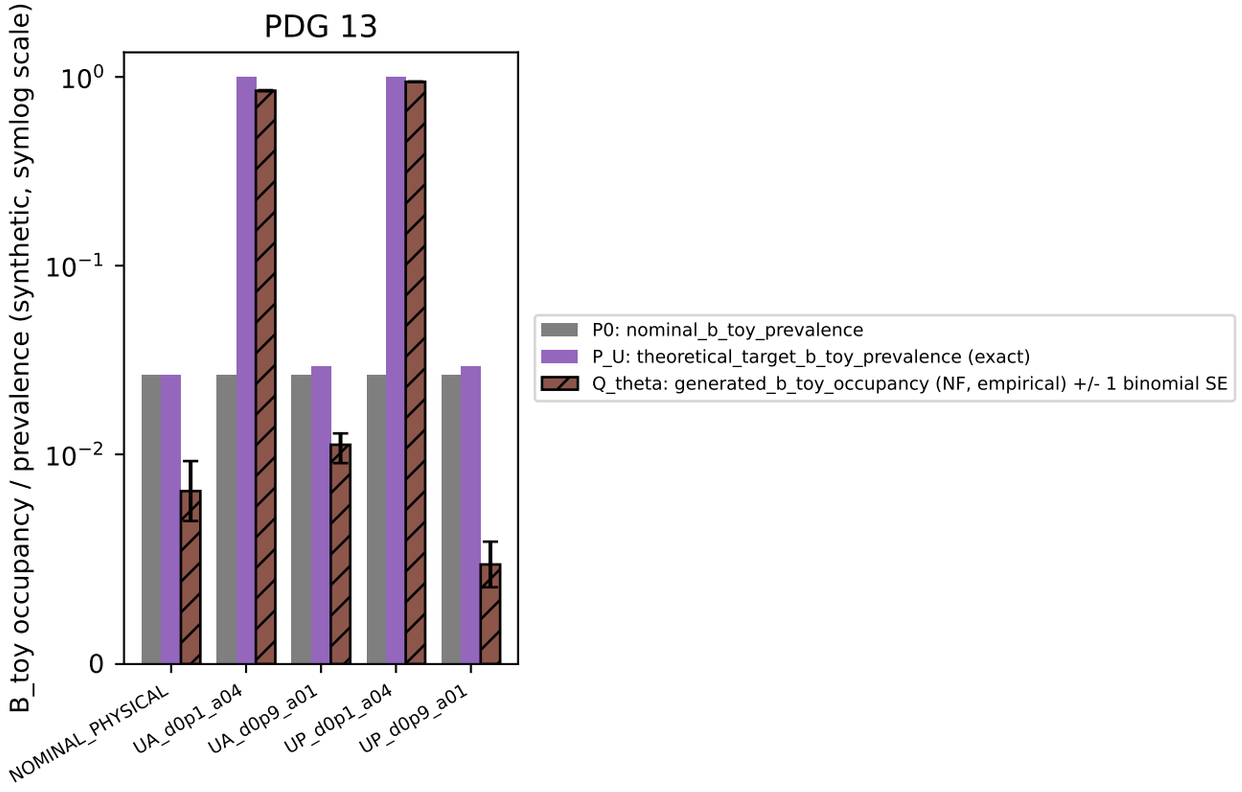
NF-generated samples versus the exact declared tilt target, per trained variant (synthetic B_toy; NLL annotations are a separate, non-bar diagnostic -- not evidence of sampling agreement)

final_train_nll: NOMINAL_PHYSICAL=2.14, UA_d0p1_a04=0.80, UA_d0p9_a01=2.62, UP_d0p1_a04=0.06, UP_d0p9_a01=2.36



final_train_nll: NOMINAL_PHYSICAL=2.56, UA_d0p1_a04=1.94, UA_d0p9_a01=2.63, UP_d0p1_a04=1.18, UP_d0p9_a01=2.72



Bounded CPU-scale affine-coupling flow (8 blocks, width 96, <=40 epochs, early stopping), not a converged production model. generated_b_toy_occupancy is measured from a small sample (see generated_sample_count per run, typically 4000) drawn from the trained flow and mapped back to physical space. For the strong tilts it approaches but does not exactly reach the declared theoretical_target_b_toy_prevalence, which is expected at this training budget. For the mild tilts, the occupancy measurement is not reliably distinguishable from sampling noise at this sample budget (binomial standard error is a sizeable fraction of the signal) and does not consistently track the tilt's declared direction -- this is a stated NF-training-budget limitation, not evidence of a sampler or mathematics bug (those are validated separately and exactly via the closed-form/direct-summation/alias-sampler path). Note: the y-axis is symlog-scaled to keep the small mild-tilt bars legible alongside the near-1.0 strong-tilt bars -- bar heights are therefore not linearly proportional to value; compare the printed numbers, not the visual bar-height ratios.